Spectral–Spatial Dual Graph Unfolding Network for Multispectral and Hyperspectral Image Fusion

Kai Zhang, *Member, IEEE*, Jun Yan, Feng Zhang, Chiru Ge, Wenbo Wan, Jiande Sun, and Huaxiang Zhang

***Abstract*— Recently, deep neural network (DNN)-based methods have achieved good results in terms of the fusion of low-spatial-resolution hyperspectral (LR HS) and high-spatial-resolution multispectral (HR MS) images. However, the spectral band correlation (SBC) and the spatial nonlocal similarity (SNS) in hyperspectral (HS) images are not sufficiently exploited by them. To model the two priors efficiently, we propose a spectral–spatial dual graph unfolding network (SDGU-Net), which is derived from the optimization of graph regularized restoration models. Specifically, we introduce spectral and spatial graphs to regularize the reconstruction of the desired high-spatial-resolution HS (HR HS) image. To explore the SBC and SNS priors of HS images in feature space and utilize the powerful learning ability of DNNs simultaneously, the iterative optimization of the spectral and spatial graph regularized models is unfolded as a network, which is composed of spectral and spatial graph unfolding modules. The two kinds of modules are designed according to the solutions of the spectral and spatial graph regularized models. In these modules, we employ graph convolution networks (GCNs) to capture the SBC and SNS in the fused image. Then, the learned features are integrated by the corresponding feature fusion modules and fed into the feature condense module to generate the HR HS image. We conduct extensive experiments on three benchmark datasets, and the results demonstrate the effectiveness of our proposed SDGU-Net.**

***Index Terms*— Graph unfolding network, hyperspectral (HS) image, image fusion, multispectral (MS) image, remote sensing.**

I. Introduction

HYPERSPECTRAL (HS) imaging technology can acquire tens or hundreds of spectral bands to describe the observed scenes, which has gradually become one of the most vital achievements in the field of remote sensing [1]. The obtained HS image not only records the spatial information of these scenes but also describes the spectral radiation information along the wavelength. Because of the abundant spectral information, the objects in the observed scenes are able to be identified by HS images, which have been applied to many fields, such as classification [2], detection [3], and mapping [4]. However, considering the physical limitations of remote sensing imaging systems, it is impractical to ensure that the HS images have both high spectral and spatial resolutions at the same time [5]. As an alternative solution, low-spatial-resolution HS (LR HS) images and high-spatial-resolution multispectral (HR MS) images are fused to generate high-spatial-resolution HS (HR HS) images [6], [7].

Over the last decades, many methods fusing LR HS and HR MS images have been proposed and produced good fusion results [8]. For example, some fusion methods [9], [10] for multispectral (MS) and panchromatic images are extended to integrate LR HS and HR MS images. Some methods [11] are developed by embedding the spectral and spatial degradation models into an unmixing framework to synthesize the desired HR HS images. In recent years, the development of deep neural networks (DNNs) has further boosted the quality of HR HS images [12]. For instance, various DNNs with different architectures are designed to learn the spectral and spatial information in HS images [13], [14]. Although satisfactory fused images are produced, these methods have not yet fully exploited the abundant priors in HS images, especially the spectral band correlation (SBC) and the spatial nonlocal similarity (SNS), to constrain the synthesis of HR HS images.

Since the high-dimensional properties of HS images, there are various priors that are helpful for their reconstruction [15]. The most well-known priors are SBC and SNS. SBC benefits from the high spectral resolution of HS images, and adjacent bands in these images are often highly correlated. SNS refers to that similar structures recur in images frequently, and the target structure can be approximated well by similar ones from other places. As an effective tool for modeling similarity relationships, graph representation has been applied to capture the SBC and SNS in images for the fusion of LR HS and HR MS images. Zhang et al. [16] employed two graphs derived from LR HS and HR MS images to regularize the restoration model. Bu et al. [17] also devised a graph Laplacian regularizer to describe the similarities along spectral and spatial dimensions. However, to reduce the computational complexity, these methods constructed spectral and spatial graphs within a local window or handle the spatial dimensions separately. Therefore, they ignore the structural similarity (SSIM) among the nodes from the whole image. Besides, the SBC and SNS in deep feature space are still not explored sufficiently by

Manuscript received 1 September 2023; revised 16 November 2023 and 7 January 2024; accepted 9 February 2024. Date of publication 13 February 2024; date of current version 20 February 2024. This work was supported in part by the Natural Science Foundation of China under Grant 61901246, in part by the China Postdoctoral Science Foundation under Grant 2019TQ0190 and Grant 2019M662432, in part by the Scientific Research Leader Studio of Jinan under Grant 2021GXRC081, and in part by the Joint Project for Smart Computing of Shandong Natural Science Foundation under Grant ZR2020LZH015. *(Corresponding author: Jiande Sun.)*

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Digital Object Identifier 10.1109/TGRS.2024.3365719

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graph representation, which limits the quality improvement of the fused images. Recently, graph convolutional networks (GCNs) have been used in image fusion tasks to capture spatial and spectral relationships in fused images. For example, Xing et al. [18] used GCN for the task of MS and panchromatic image fusion to refine the spectral relationships between different bands in HR MS images. Meanwhile, Yan et al. [19] designed a spatial GCN module and a spectral GCN module to explore the long-range spatial and spectral relationships for pansharpening. However, exploring SBC and SNS using GCNs is still not effectively utilized in the fusion of LR HS and HR MS images.

To address the aforementioned issues, we propose a novel network, dubbed as spectral–spatial dual graph unfolding network (SDGU-Net), by unfolding the spectral–spatial dual graph regularized restoration model between the fused image and source images. Specifically, we reformulate the spectral and spatial graph regularized restoration models and use the proximal gradient descent [20] to iteratively solve them. Inspired by the model-driven framework, we unfold the iterative algorithms and establish spectral and spatial graph unfolding modules to achieve their optimization in the form of a network. In the spectral graph unfolding module, a GCN is considered to model the SBC in feature space. Similarly, the input of the spatial graph unfolding module is segmented into patches, and a GCN is imposed on them to learn the SNS in images globally. Then, the features extracted by spectral and spatial graph unfolding modules are integrated by feature fusion modules, which further enhance the spectral and spatial information in the features from different modules. Finally, the fused information is combined by a feature condense module for the reconstruction of the desired HR HS image. To the best of our knowledge, this is the first time that spectral and spatial regularization optimization models are unfolded as a network for the fusion of LR HS and HR MS images. Experiments on four benchmark datasets illustrate that the proposed SDGU-Net obtains better qualitative and quantitative results compared with some state-of-the-art counterparts. The contributions of our proposed SDGU-Net are summarized as follows.

1) We propose iterative algorithms based on the proximal gradient descent to optimize the spectral and spatial graph regularized restoration models for the generation of the fused image.
2) We unfold the iterative algorithms as a network consisting of spectral and spatial graph unfolding modules. The two modules are responsible for the learning of the SBC and SNS in feature space by GCNs.
3) We introduce feature fusion and condense modules to merge the spectral and spatial information in the features from different modules more efficiently.

The rest of this article is organized as follows. Section II introduces some methods based on traditional paradigms and DNNs. In Section III, spectral and spatial regularization restoration models are formulated and optimized. The proposed SDGU-Net is detailed in Section IV. In Section V, fusion experiments on four datasets and ablation experiments are performed to verify the effectiveness of the proposed SDGU-Net. Section VI provides the conclusions.

II. Related Work

A. Traditional Methods

With the developments of imaging sensors, more and more HS images are obtained, and the image fusion technique is also attracting increasing attention to improve their resolutions. Owing to the similarity in terms of formulation, some methods for the fusion of MS and panchromatic images are further generalized to merge LR HS and HR MS images. For example, Chen et al. [21] divided the bands in LR HS images into several groups and used the synthesized bands in HR MS images to enhance the resolutions of these groups. Inspired by the intrinsic properties of HS images, spectral unmixing is combined with the restoration models to generate the fused HR HS images. Given the ill-posedness of the restoration models, researchers adopted distinct priors to constrain these models and recovered the fused images better. Simoes et al. [22] utilized the vector total variation to preserve the edges in the HR HS images. Following a similar framework, sparsity [23], [24] is introduced to estimate more informative endmembers. Moreover, the unmixing-based fusion models were reformulated into a Bayesian framework and solved by the maximum a posteriori [25], [26]. Besides, Zhang et al. [27] incorporated robust principal component analysis [28] with the spectral degradation model, in which the SBC in HS images was characterized by a low-rank constraint.

To avoid the structural damage of high-dimensional data, the fusion of LR HS and HR MS images is mathematically extended to the tensor version. For instance, Xu et al. [29] proposed a tensor canonical polyadic decomposition, in which similar cubes in images were aggregated as a tensor. Borsoi et al. [30] considered Tucker decomposition to model the structures in HS images along spectral and spatial directions. Zare et al. [31] further introduced the nonnegativity of pixel values into tensor decomposition to provide a stricter constraint. Because most of these methods are regularized by priors in HS images, the modeling of these priors has a large influence on the reconstruction of the fused images. Due to the iterative optimization algorithms, traditional methods generally suffer from high-computational complexity.

B. DNN-Based Methods

In recent years, researchers utilize the powerful capabilities of the nonlinear representation of DNNs to learn the complex mapping between the fused images and source images. As one typical structure of DNNs, the convolution neural network (CNN) is first applied to the fusion of LR HS and HR MS images [32], [33], [34], [35]. Considering the high-dimensional attributes of HS images, Palsson et al. [36] employed a 3-D CNN to extract the spectral and spatial features in HS images. Under the variational inference framework, Wang et al. [37] designed two networks for the latent representation and the spectral mixing model of HS images, respectively. Li et al. [38] proposed three subnetworks for feature extraction, in which the learning ability of degradations

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between the fused images and source images was enhanced. Subsequently, various DNNs are introduced into this task. For example, the residual network was also taken as an extractor for the learning of high-frequency information in images [39]. Some researchers [40], [41], [42] proposed attention modules to describe complex mapping relationships more efficiently. Recently, Gao et al. [43] combined attention mechanism with degradation estimation and extracted spatial and spectral features through a designed autoencoder. Besides, generative adversarial networks, recurrent neural networks, and invertible neural networks were also considered in [44], [45], [46], and [47] to produce better results. Very recently, Jia et al. [48] proposed a network based on the transformer to enhance the efficiency of modeling long-distance dependence as well as to capture features with different resolutions.

Inspired by the iterative optimization in traditional methods, unfolding-based DNNs [49], [50], [51], [52], [53], [54] are drawing more attention due to their decent properties and better interpretability. For instance, Yang et al. [54] unfolded the half-quadratic splitting algorithm and constructed a variational regularization network. In addition, Xie et al. [55] developed an unfolding-based DNN according to the proximal gradient algorithm, in which the low-rank prior was cast on the deep features to obtain a more concise representation. Sun et al. [56] also derived a network from the backprojection constrained by low-rank priors. Khader et al. [57] imposed the nonnegativity and sparsity on the unfolding network to further enhance the information in HS images. Furthermore, Khader et al. [58] proposed a model-guided unfolding network by considering the nonlocal spatial and spectral priors. In this method, the network is constructed by unfolding the iterative backprojection. Instead of explicitly combining the nonlocal self-similarity (NLSS) with the physical model, NLSS modules were heuristically designed and embedded into the unfolding network. On the contrary, the proposed SDGU-Net is established by unfolding the proximal gradient descent optimization of the graph regularized restoration model. The graph unfolding modules in SDGU-Net are directly devised according to the updates in the iterative algorithm. Considering the internal connection between GCNs and the graph constraint in our proposed fusion model, we adopt GCNs to achieve the learning of graph Laplacian regularization in spatial and spectral domains. Compared with the self-attention in transformer-based methods [48], GCNs are more closely connected with the graph Laplacian regularization and are more similar in form. On the one hand, the proposed SDGU-Net is constructed according to the optimization of the spectral and spatial graph regularized restoration models. On the other hand, the graph Laplacian regularization in SDGU-Net is achieved by GCNs. So, the proposed SDGU-Net has good interpretability.

III. Spectral–Spatial Regularized Optimization

From the perspective of image restoration, LR HS and HR MS images are viewed as the degradations of the desired HR HS image in spatial and spectral domains. Then, these relationships are regularized by various priors and solved to produce the fused image. In this article, the restoration models are constrained by spectral and spatial graphs, which are used to capture the SBC and SNS in the fused image. Their formulations and optimizations are detailed as follows.

A. Spectral Graph Regularized Optimization

Let $\mathcal{X} \in \mathbb{R}^{W\times H\times B}$ and $\mathcal{Y} \in \mathbb{R}^{W\times H\times B}$ denote the fused image and the upsampled LR HS image, respectively. W and H are the spatial sizes of the corresponding images. B stands for the number of bands in HS images. Then, the relationship between $\mathcal{X}$ and $\mathcal{Y}$ can be defined as follows:

$$\mathbf{Y} = \mathbf{D}\tilde{X} + n_1 \tag{1}$$

where $\tilde{\mathbf{X}} \in \mathbb{R}^{WH\times B}$ and $\mathbf{Y} \in \mathbb{R}^{WH\times B}$ are the matrix versions of $\mathcal{X}$ and $\mathcal{Y}$, respectively. The spatial blurring is denoted by $\mathbf{D} \in \mathbb{R}^{WH\times WH}$. n_1 is the noise.

To preserve the high SBC in the fused image, we construct a spectral graph with the Laplacian matrix $\mathbf{L}_1 \in \mathbb{R}^{B\times B}$, in which the bands of the fused image are regarded as nodes, and the elements in $\mathbf{L}_1$ denote the correlation between nodes. Then, introducing the graph regularization in the spectral domain, the reconstruction of the fused image is achieved by minimizing the following function:

$$\tilde{\mathbf{X}} = \arg\min_{\tilde{\mathbf{X}}} \frac{1}{2}\left\|\mathbf{Y} - \mathbf{D}\tilde{X}\right\|_F^2 + \frac{\alpha}{2}\mathrm{tr}\left(\tilde{\mathbf{X}}\mathbf{L}_1\tilde{\mathbf{X}}^T\right) \tag{2}$$

where $\mathrm{tr}(\cdot)$ denote the trace operation and α is a trade-off parameter. Because the correlation among different bands is recorded in $\mathbf{L}_1$, the SBC in the fused image will be ensured through the minimization of (2).

Considering that (2) is the form of $f(\mathbf{X}) + g(\mathbf{X})$, (2) is optimized by the proximal gradient descent method [20]. Although other optimization algorithms can also solve this problem, such as the alternating direction method of multipliers [52] and half-quadratic splitting [54], the proximal gradient descent method does not require the introduction of additional auxiliary variables and will be more concise when unfolding into a network. Thus, we lead to the gradient descent of $f(\cdot)$ and the proximal mapping of $g(\cdot)$

$$\begin{cases} \mathbf{Q}_i = \tilde{\mathbf{X}}_i - \lambda\nabla f = \left(\mathbf{I} - \lambda\mathbf{D}^T\mathbf{D}\right)\tilde{\mathbf{X}}_i + \lambda\mathbf{D}^T\mathbf{Y} \\ \tilde{\mathbf{X}}_{i+1} = \mathrm{prox}_g(\mathbf{Q}_i) \\ \qquad = \arg\min_{\mathbf{X}}\left(\|\mathbf{X} - \mathbf{Q}_i\|_F^2 + \frac{\alpha}{2}\mathrm{tr}\left(\mathbf{X}\mathbf{L}_1\mathbf{X}^T\right)\right) \end{cases} \tag{3}$$

where λ and ∇ denote the step size and the derivative operator, respectively. $\mathrm{prox}_g(\cdot)$ is the proximal operator. i is the number of iterations. By solving the proximal operator, a closed-form solution can be computed as follows:

$$\tilde{\mathbf{X}}_{i+1} = \left(\left(\mathbf{I} - \lambda\mathbf{D}^T\mathbf{D}\right)\tilde{\mathbf{X}}_i + \lambda\mathbf{D}^T\mathbf{Y}\right)(\mathbf{I} + \alpha\mathbf{L}_1)^{-1}. \tag{4}$$

Then, we can update $\tilde{\mathbf{X}}$ iteratively until the optimization is converged.

B. Spatial Graph Regularized Optimization

Similarly, we can write the spectral degradation between the fused image and the HR MS image $\mathcal{Z} \in \mathbb{R}^{W\times H\times b}$ as follows:

$$\mathbf{Z} = \mathbf{S}\hat{\mathbf{X}} + n_2 \tag{5}$$

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where $\mathbf{Z} \in \mathbb{R}^{p^2b\times N}$ is obtained by partitioning $\mathcal{Z}$ into overlapped patches with the size of $p \times p$ and rearranging these patches as the columns in $\mathbf{Z}$. The overlap size is one half of the patch size. In the same way, $\mathcal{X}$ is rearranged as $\hat{\mathbf{X}} \in \mathbb{R}^{p^2B\times N}$. b is the number of bands of the HR MS image. The number of patches is denoted by N. $\mathbf{S} \in \mathbb{R}^{p^2b\times p^2B}$ is the spectral degradation matrix, and n_2 is the additive noise.

In (5), each column in $\hat{\mathbf{X}}$ or $\mathbf{Z}$ is one local patch of the corresponding image. Therefore, it is straightforward to seamlessly embed the SNS prior into the restoration of $\hat{\mathbf{X}}$. Specifically, a spatial graph Laplacian matrix $\mathbf{L}_2 \in \mathbb{R}^{N\times N}$ is constructed. Each patch in $\mathcal{X}$ is thought of as one node, and the similarity between these patches is encoded into $\mathbf{L}_2$. So, the spatial graph regularized function can be formulated as follows:

$$\hat{\mathbf{X}} = \arg\min_{\hat{\mathbf{X}}} \frac{1}{2}\|\mathbf{Z} - \mathbf{S}\hat{\mathbf{X}}\|_F^2 + \frac{\beta}{2}\mathrm{tr}\left(\hat{\mathbf{X}}\mathbf{L}_2\hat{\mathbf{X}}^T\right) \tag{6}$$

where β is a trade-off parameter. In (6), the spatial graph Laplacian regularizer facilitates the preservation of the SNS in $\mathcal{X}$.

Then, according to the proximal gradient descent method, $\hat{\mathbf{X}}$ is derived by updating the following equation:

$$\hat{\mathbf{X}}_{i+1} = \left(\left(\mathbf{I} - \lambda\mathbf{S}^T\mathbf{S}\right)\hat{\mathbf{X}}_i + \lambda\mathbf{S}^T\mathbf{Z}\right)(\mathbf{I} + \beta\mathbf{L}_2)^{-1}. \tag{7}$$

Although $\mathcal{X}$ can be obtained by updating (4) and (7), the representation abilities of $\mathbf{L}_1$ and $\mathbf{L}_2$ are limited, because they are estimated in the original image domain. Moreover, some methods [16], [59] construct $\mathbf{L}_1$ and $\mathbf{L}_2$ within a local region, which cannot efficiently capture the global SBC and SNS priors in the fused images. To improve the performance of spectral and spatial regularization optimization models, we unfold (4) and (7) and propose SDGU-Net, in which GCNs are adopted to globally infer the neighbors of different nodes in feature space. The details of the proposed SDGU-Net are presented in Section IV.

IV. SDGU-Net

Fig. 1 illustrates the architecture of the proposed SDGU-Net, which is established by unfolding the iterative update of (4) and (7). First, the upsampled LR HS image and the HR MS image are processed by a convolution layer to project them into the feature space. Then, according to the unfolding optimization of (4), a spectral GCN and a spectral graph unfolding module are constructed, which models the SBC in the fused image. We also design a spatial GCN and a spatial graph unfolding module from (7) in the same manner. They are responsible for the learning of the SNS in the desired HR HS image. Next, spectral and spatial graph unfolding modules are both repeated K stages to exploit the SBC and SNS in deep feature space. Finally, the features from the spectral and spatial graph unfolding modules at all stages are integrated by the corresponding feature fusion modules, and their outputs are fed into the feature condense module for the reconstruction of the fused HR HS image. Through the unfolding of spectral and spatial graph regularized optimization, the SBC and SNS in the fused image are preserved better, and the quality of the fused result is improved. More technical details are introduced in the following parts.

A. Spectral Graph Unfolding Module

1) Unfolding Spectral Graph Optimization: To improve the ability of feature extraction, residual learning [60] is considered in the proposed spectral graph unfolding module. Specifically, we define the residual as $e_i = \tilde{\mathbf{X}}_0 - \tilde{\mathbf{X}}_i$, where $\tilde{\mathbf{X}}_0 = (\mathbf{D}^T\mathbf{D})^{-1}\mathbf{D}^T\mathbf{Y}$ and is the least-squares solution of the first term in (2). Then, substituting $\tilde{\mathbf{X}}_{i+1}$ and $\tilde{\mathbf{X}}_i$ in (4), it is simplified as follows:

$$e_{i+1} = \left(\left(\mathbf{I} - \lambda\mathbf{D}^T\mathbf{D}\right)e_i + \alpha\tilde{\mathbf{X}}_0\mathbf{L}_1\right)(\mathbf{I} + \alpha\mathbf{L}_1)^{-1}. \tag{8}$$

Then, (8) is unfolded as the structure in Fig. 2(a). In Fig. 2, $\mathbf{I} - \lambda\mathbf{D}^T\mathbf{D}$ is defined as a spatial residual block. In $\mathbf{I} - \lambda\mathbf{D}^T\mathbf{D}$, e_i is degraded by $\mathbf{D}$ and then improved by its inverse operation $\mathbf{D}^T$. The error is obtained from the difference between e_i and its spatial degradation version. For $\alpha\tilde{\mathbf{X}}_0\mathbf{L}_1$, $\mathbf{L}_1$ encodes the correlation among spectral bands, and $\tilde{\mathbf{X}}_0\mathbf{L}_1$ can be viewed as a graph convolution. So, a spectral GCN is proposed to explore the correlation between different channels in feature space. For $(\mathbf{I} + \alpha\mathbf{L}_1)^{-1}$, it contains the spectral graph Laplacian matrix $\mathbf{L}_1$. Moreover, $e_i^m(\mathbf{I} + \alpha\mathbf{L}_1)^{-1}$ has the same form as $\tilde{\mathbf{X}}_0\mathbf{L}_1$, where e_i^m is the intermediate result and equal to $(\mathbf{I} - \lambda\mathbf{D}^T\mathbf{D})e_i + \alpha\tilde{\mathbf{X}}_0\mathbf{L}_1$. Similarly, we also use a spectral GCN to learn the correlation relationships in e_i^m. Because of the inverse operation on $\mathbf{I}+\alpha\mathbf{L}_1$, the network for $(\mathbf{I} + \alpha\mathbf{L}_1)^{-1}$ is named the inverse spectral GCN.

2) Module Structure: According to the above analysis, we further enrich the structural details of the proposed spectral graph unfolding module, as shown in Fig. 2(b). In the spatial residual block, a strided convolution (SConv) layer and a deconvolution (DConv) layer are introduced to learn the operations of $\mathbf{D}$ and $\mathbf{D}^T$, respectively. In the two layers, 64 filters with the size of 2×2 are used for feature extraction, and the stride is set as 2. Then, the residual is obtained by computing the difference between e_i and its corresponding degraded result.

To use GCN, we need to construct graph connections and graph nodes. In the inverse spectral GCN, the intermediate result e_i^m is downsampled by two average pooling with a ratio of t. Then, we can obtain two feature maps, $E_i^1 \in \mathbb{R}^{W/t\times H/t\times C}$ and $E_i^2 \in \mathbb{R}^{W/t\times H/t\times C}$, which are then reshaped as matrix form to calculate the graph connectivity in GCN. C is the number of channels. Here, we regard each channel in $E_i^3 \in \mathbb{R}^{W\times H\times C}$ as the graph node. The similarity between nodes is recorded by the elements in $\mathbf{A}_1 \in \mathbb{R}^{C\times C}$, which is estimated from the multiplication between matrix forms of E_i^1 and E_i^2. In detail, $\mathbf{A}_1$ is learned and progressively optimized through the network to act as graph connections and is considered as the corresponding version of $\mathbf{L}_1$. Then, $\mathbf{A}_1$ is normalized by softmax and incorporated with $E_i^3 \in \mathbb{R}^{W/t\times H/t\times C}$. The formulation is written as follows:

$$e_{i+1} = \mathcal{R}^{-1}\left(\mathcal{R}\left(E_i^3\right)\mathrm{Softmax}(\mathbf{A}_1)\right) \tag{9}$$

where $\mathcal{R}(\cdot)$ and $\mathcal{R}^{-1}(\cdot)$ are the reshaping operator and its inverse operation, respectively. In the inverse spectral GCN, the similarity between channels of feature maps is computed and further used to enhance the correlation of the channels in E_i^3. Through the formulation, the SBC in the fused image

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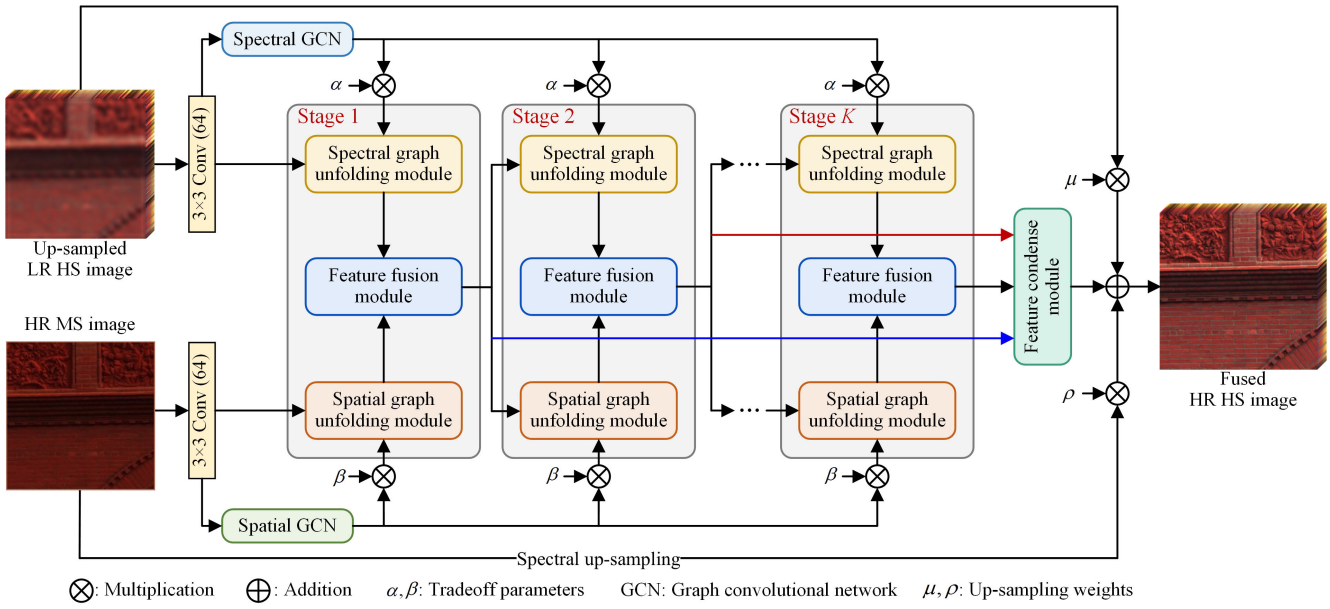


Fig. 1. Illustration of the proposed SDGU-Net.

is preserved better. Because of the similar formulation, the spectral GCN for $\tilde{\mathbf{X}}_0$ and the inverse spectral GCN for e_i^m share the same structure to depict the spectral information in images.

B. Spatial Graph Unfolding Module

1) Unfolding Spatial Graph Optimization: Through the way to unfold the optimization of (6), the residual in the spatial graph unfolding module is defined as $d_i = \hat{\mathbf{X}}_0 - \hat{\mathbf{X}}_i$, where $\hat{\mathbf{X}}_0 = (\mathbf{S}^T\mathbf{S})^{-1}\mathbf{S}^T\mathbf{Z}$. Then, (7) is reduced to

$$d_{i+1} = \left(\left(\mathbf{I} - \lambda\mathbf{S}^T\mathbf{S}\right)d_i + \beta\hat{\mathbf{X}}_0\mathbf{L}_2\right)(\mathbf{I} + \beta\mathbf{L}_2)^{-1}. \tag{10}$$

Fig. 3(a) shows the unfolded structure of (10). In Fig. 3(a), we consider $\mathbf{I} - \lambda\mathbf{S}^T\mathbf{S}$ as a spectral residual block, because it calculates the difference between d_i and its degraded result in the spectral domain. Similar to the modules in Fig. 2(a), $\mathbf{L}_2$ and $(\mathbf{I} + \beta\mathbf{L}_2)^{-1}$ are achieved by a spatial GCN and an inverse spatial GCN, respectively. By the two modules, the SNS in the fused image is retained better.

2) Module Structure: The proposed spatial graph unfolding module is composed of a spectral residual block and an inverse spatial GCN. In the spectral residual block, the number of channels of feature maps is adjusted by two 1×1 convolution layers to simulate the operations of $\mathbf{S}$ and $\mathbf{S}^T$ along the spectral dimension. Then, d_i is subtracted by its degraded result to produce the residual. As the sum of the residual and the output of the spectral GCN, the intermediate result d_i^m flows into the following inverse spatial GCN.

In the inverse spatial GCN, the construction of graph nodes and graph connections is also required. d_i^m is projected by two convolution layers containing 64 3×3 filters. We segment the outputs of the two layers into $p \times p$ patches with half overlapping and flatten them as matrices $D_i^1 \in \mathbb{R}^{p^2C\times N}$ and $D_i^2 \in \mathbb{R}^{p^2C\times N}$. In the same way, we can obtain D_i^3, in which each column is viewed as a graph node. Then, graph connectivity is obtained by estimating the similarity between the columns from D_i^1 and D_i^2. To further exploit the similarity between different patches, a fully connected (FC) layer is implemented on each column of D_i^1 and D_i^2. Then, the outputs of the two FC layers are multiplied for the calculation of the similarity matrix $\mathbf{A}_2$, which contains the pairwise similarity of different patches in the whole feature maps. Specifically, we similarly consider $\mathbf{A}_2$ as the counterpart of $\mathbf{L}_2$ and utilize it to obtain graph connections in the inverse spatial GCN. To embed the SNS encoded by $\mathbf{A}_2$ into the fused image, $\mathbf{A}_2$ is combined with D_i^3. Finally, all columns in the product are aggregated to synthesize the enhanced feature d_{i+1}. Mathematically, the enhancement of d_i^m is summarized as follows:

$$d_{i+1} = \mathcal{A}\left(D_i^3\text{Softmax}\left(\text{FC}\left(D_i^1\right)^T\text{FC}\left(D_i^2\right)\right)\right) \tag{11}$$

where $\mathcal{A}(\cdot)$ is the aggregation operator. In the inverse spatial GCN, we use the similarity matrix of all patches to promote the SNS in feature space. Because all patches in the whole image are considered, the SNS in the fused image is utilized more sufficiently. Thus, the spatial structures in the fused image can be enhanced better. Considering the similar form of the spatial GCN, we employ the same structure as that of the inverse spatial GCN to learn the SNS in $\hat{\mathbf{X}}_0$.

C. Reconstruction of the Fused Image

After the enhancement by spectral and spatial graph unfolding modules, the SBC and SNS are inherited into the features. Then, these features are merged by the corresponding feature fusion module, whose structure is shown in Fig. 4. In this module, e_i and d_i are first averaged to integrate the captured spectral and spatial information. Specifically, e_i and d_i are both multiplied by 0.5 and added, which allows more spectral and spatial details to be extracted from e_i and d_i. In this process, all operations keep the size constant. As shown in Fig. 4,

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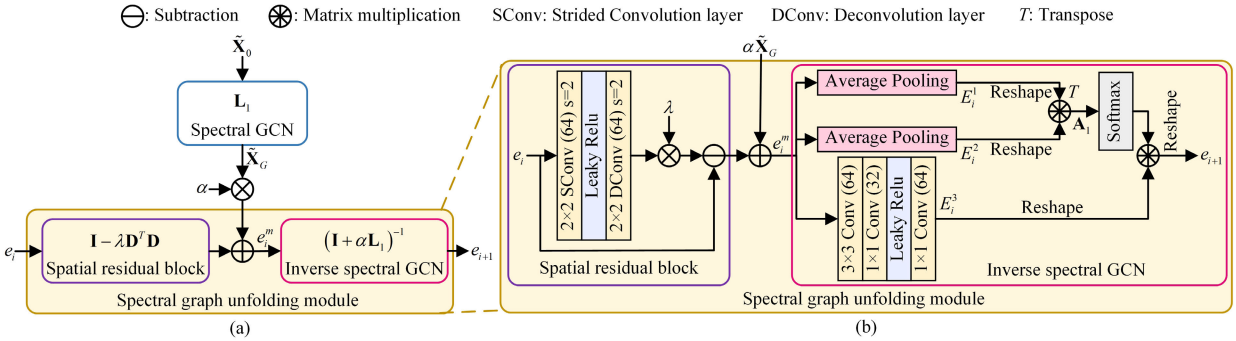


Fig. 2. Spectral graph unfolding module derived from (8). (a) Spectral graph unfolding module. (b) Details of spectral graph unfolding module.

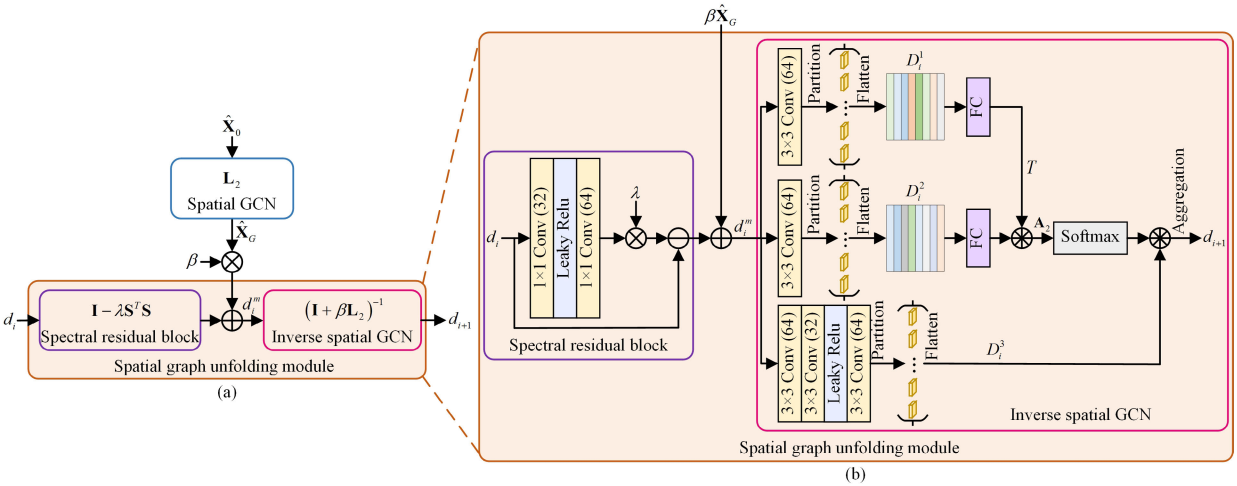


Fig. 3. Spatial graph unfolding module derived from (10). (a) Spatial graph unfolding module. (b) Details of spatial graph unfolding module.

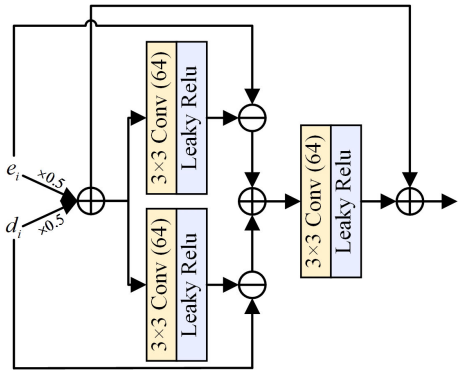


Fig. 4. Structure of the feature fusion module.

to compensate for the information loss in the fusion process, we further estimate the details based on the difference between the average results and its projected features. Then, the spectral and spatial distortions can be reduced. Finally, a residual connection is introduced to combine the extracted details and the averaged result for the generation of the output of the feature fusion module. Through the module, the differences between e_i and d_i are used sufficiently, and the spectral and spatial details among them are closely integrated.

When the fused features of different feature fusion modules are produced, we design a feature condense module to reduce these features to the matched dimension of the fused image. Fig. 5 shows the structure of the feature condense module. In this module, we first concatenate the outputs of all feature fusion modules along the channel dimension and process them to the matched size through a convolution layer containing 64 3×3 filters. Subsequently, an attention map M is derived by a sigmoid layer to further estimate the spectral and spatial details. In specific, the attention map and its complementary map are considered to emphasize the salient content in feature maps. Finally, the feature maps are convoluted by B filters with the size of 3×3 to obtain its output $\mathcal{O} \in \mathbb{R}^{W \times H \times B}$. So, the desired HR HS image is generated by the equation

$$\mathcal{X} = \mu \mathcal{Y} + \rho \mathcal{U}(\mathcal{Z}) + \mathcal{O} \tag{12}$$

where μ and ρ are the spectral and spatial upsampling weights, respectively. $\mathcal{U}(\cdot)$ is the spectral upsampling operator and is achieved by calculating the least-squares solution of the first term in (6).

D. Learning

Finally, our proposed SDGU-Net is trained by optimizing the following function:

$$L = \|\mathcal{G} - \mathcal{X}\|_F^2 \tag{13}$$

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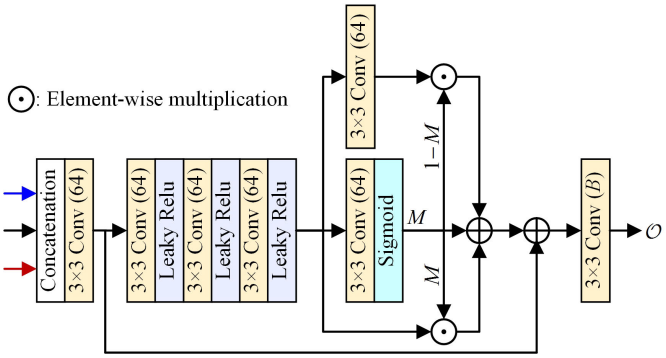


Fig. 5. Structure of the feature condense module.

where $\mathcal{G}$ is the ground truth (GT), also the reference image. Specifically, the training is implemented by the PyTorch framework on a server with Intel Core i7-9700 CPUs and an NVIDIA 2080TI GPU. We use the Adam optimizer with an initialized learning rate of 0.0001 and a batch size of 4 to minimize (13). After 2000 epochs, the learning rate is adjusted to 0.00001. The number of total epochs is 4000. Besides, we set the step size λ in (8) and (10) as 0.5. Similarly, the regularization parameters α and β are set to 0.5, and the upsampling weights μ and ρ are also set to 0.5. Their influences on fusion results are investigated in Section V.

V. Experiments

In this section, we conduct extensive experiments to illustrate the effectiveness of our proposed SDGU-Net on four benchmark datasets, including Columbia Imaging and Vision Laboratory (CAVE) [61], Harvard [62], Pavia [63], and DFC2018 Houston [64] datasets. For all these datasets, the proposed SDGU-Net adopts the same hyperparameter settings.

A. Datasets

In the four benchmark datasets mentioned above, HR HS images are provided. The CAVE dataset contains 32 512 × 512 HR HS images, and the number of spectral bands is 31. To meet the demand for the training of the compared DNN-based methods and our proposed SDGU-Net, these HR HS images are first divided into three separate parts without overlapping, taken as the training set, test set, and validation set. Subsequently, the classified images are cropped into subimages of size 256 × 256 pixels with a stride size of 64. Then, LR HS and HR MS images are produced by implementing spatial and spectral degradations on these subimages. Specifically, these subimages are blurred by applying a 4 × 4 Gaussian filter with mean 0 and standard deviation 2 and then downsampling to the size of 64 × 64 pixels for the synthesization of LR HS images. The HR MS images with the size of 256 × 256 pixels are synthesized by utilizing the spectral response function of the Nikon D700 camera [49], [50] and filtering the subimages along the spectral dimension. Finally, 238 image pairs are collected, of which 218 and 8 pairs are employed for training and validation, respectively. The original subimages are regarded as the GT. The trained models are tested on the rest of the image pairs.

The Harvard dataset is composed of 50 HR HS images with a size of 1392 × 1040 pixels and 31 spectral bands. Afterward, we divide them into three nonoverlapping parts for training, testing, and validation. For the training dataset, we set the stride size to 64, resulting in 860 subimages of size 256 × 256 pixels. Then, the same synthesis scheme as that in the CAVE dataset is imposed on the 860 subimages to generate LR HS and HR MS images to be fused. In the training dataset, the sizes of LR HS and HR MS images are 64 × 64 and 256 × 256 pixels, respectively. Similarly, the test and validation datasets are segmented into subimages of size 512 × 512 pixels, and the extracted patches have a stride size of 128. After the spatial and spectral degradation, LR HS and HR MS images are obtained with the sizes of 128 × 128 and 512 × 512 pixels, respectively. Subsequently, DNN-based methods are trained on 860 image pairs and validated on 12 image pairs. The remaining 30 pairs of images are prepared for testing.

The Pavia dataset is a remote sensing image consisting of 103 bands, 98 of which are available for experiments, because the bands with serious water vapor absorption are removed, whose size is 610 × 340 pixels. For more rational training, the dataset is partitioned into three parts that do not intersect each other, which are used as the training set, test set, and validation set. In this training dataset, 342 subimages with the size of 64 × 64 pixels are extracted using a stride size of 16 and degraded to generate LR HS and HR MS images. Similarly, LR HS images are acquired by downsampling and blurring the original subimages. The spectral response function of the QuickBird satellite [26] is combined with the subimages to synthesize the HR MS images. To prepare the test samples, we extract overlapping subimages of size 256 × 256 pixels from the test dataset as the desired HR HS images with a stride size of 64. The degradation operation is performed to obtain LR HS and HR MS images of size 64 × 64 and 256 × 256 pixels. The validation dataset is collected in the same way as that used for the test dataset. Then, we train and validate DNN-based methods on 342 and 1 image pairs, respectively. Testing is conducted on the remaining ten image pairs.

The DFC2018 Houston dataset is 4172 ×1202 pixels in size and includes 48 spectral bands. To prepare the training set, test set, and validation set, we crop it into three independent parts. The size of the subimages in the training dataset is segmented to 256 × 256 pixels, and the stride size is 64. In the test and validation datasets, the size of the acquired subimages is also 256 × 256 pixels, but the stride size is set to 256. Subsequently, we perform the same spatial and spectral degradation operations on these HR HS images as on the Pavia dataset to generate the corresponding LR HS and HR MS images. Thus, we can obtain LR HS images of size 64 × 64 pixels and HR MS images of size 256 × 256 pixels. Ultimately, we use 690, 13, and 3 image pairs for training, testing, and validation, respectively.

B. Compared Methods and Metrics

1) Compared Methods: In the experimental section, our proposed SDGU-Net is compared with four methods belonging to traditional paradigms and five DNN-based methods.

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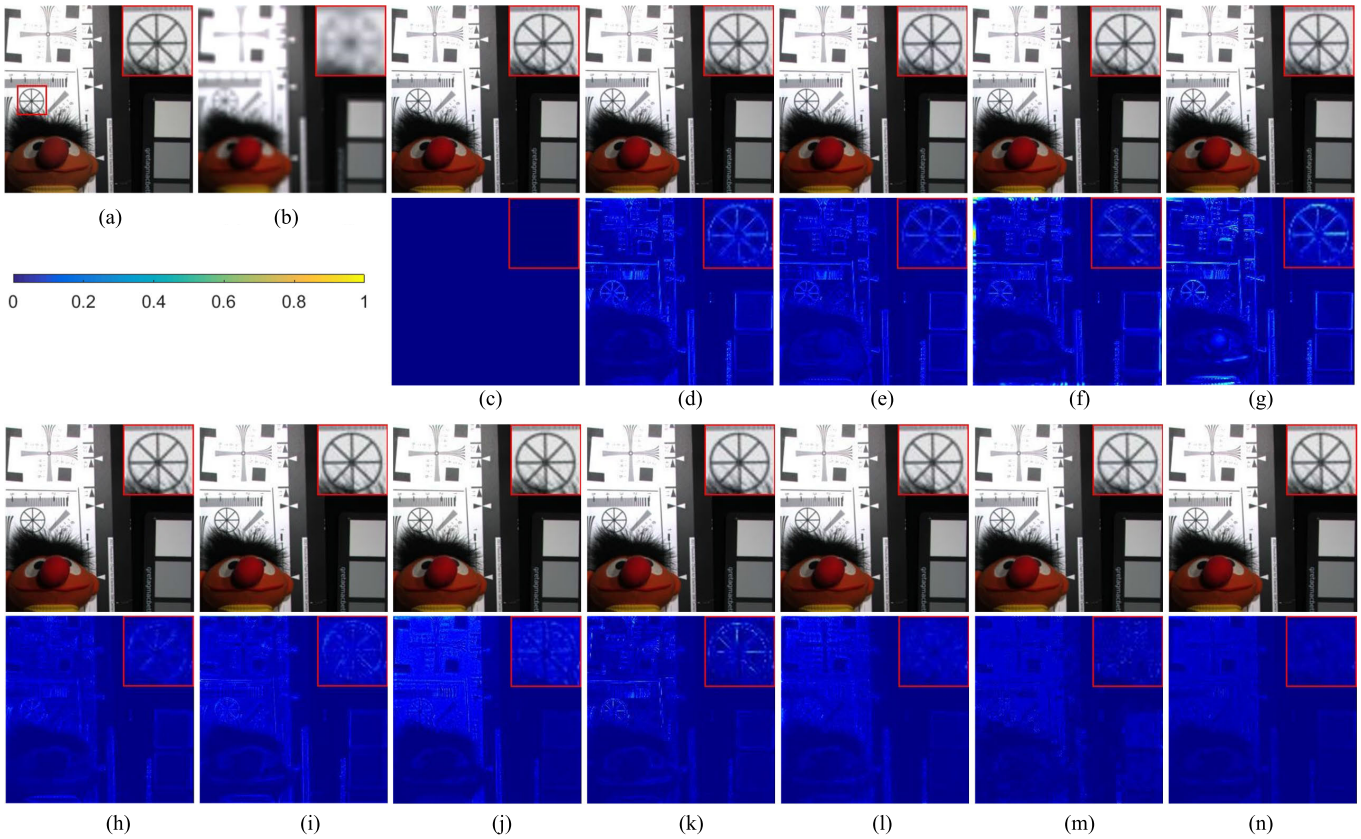


Fig. 6. Fusion results of different methods on the CAVE dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) CNMF. (e) HySure. (f) GLP-HS. (g) CSU. (h) SpfNet. (i) SSR-Net. (j) MHF-Net. (k) PSRT. (l) MSST-Net. (m) 3DT-Net. (n) Proposed SDGU-Net.

The former four methods are coupled nonnegative matrix factorization (CNMF) [11], hyperspectral superresolution (HySure) [22], generalized Laplacian (GLP-HS) [10], and coupled spectral unmixing (CSU) [23]. The latter ones are subspace-level fusion network (SpfNet) [53], spatial–spectral reconstruction network (SSR-Net) [13], multispectral and hyperspectral fusion net (MHF-Net) [55], pyramid shuffle-and-reshuffle transformer (PSRT) [14], multiscale spatial–spectral transformer network (MSST-Net) [48], and 3-D CNN and transformer prior network (3DT-Net) [65].

2) Metrics: We utilize peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM) [66], universal image quality index (UIQI) [67], SSIM [68], and relative dimensionless global error in synthesis (ERGAS) [69] to evaluate the quality of the fused HR HS images. Higher PSNR, UIQI, and SSIM values indicate better fusion performance. For SAM and ERGAS, smaller values represent that the desired results are reconstructed better.

C. Results on the CAVE Dataset

This part demonstrates the performance of our proposed SDGU-Net and the nine compared methods. The first and third rows in Fig. 6 show the visual results of one LR HS and HR MS image pair. These visual results are pseudo-color images, which are composed of the 24th, 15th, and 7th bands in the fused HR HS images. In Fig. 6, an interesting region is zoomed in and outlined by a red box for an intuitive demonstration. To further display the approximation performance, absolute error maps between GT and the fused images are also provided in the second and fourth rows in Fig. 6. It can be seen that all fusion methods improve the spatial details of HS images when compared with the LR HS image. The visual performance is close, but some differences can be found in the absolute error maps. For traditional methods, their fusion results suffer from the loss of spatial details. From their error maps, one can find that the regions containing edges or textures are not reconstructed well. For DNN-based methods, some errors also can be observed in the results of SpfNet and SSR-Net. The 3DT-Net achieves comparable results to SDGU-Net, but some reconstruction errors can be still found. Compared with other DNN-based methods, our proposed SDGU-Net can have better reconstruction performance, which is confirmed in the error map.

Table I presents the quantitative evaluations of all methods on the CAVE datasets. In this table, the metric values are averaged on all test image pairs, and the best values are highlighted in bold. The values in Table I show that the proposed SDGU-Net achieves the best results for all four metrics. At the same time, 3DT-Net obtains the best performance in SAM and UIQI metrics. Moreover, GLP-HS uses the injection of spatial details to obtain reconstructed images, which gives the best results among traditional methods. The better fusion performance of SDGU-Net benefits from the regularization of spectral and spatial graphs in the network.

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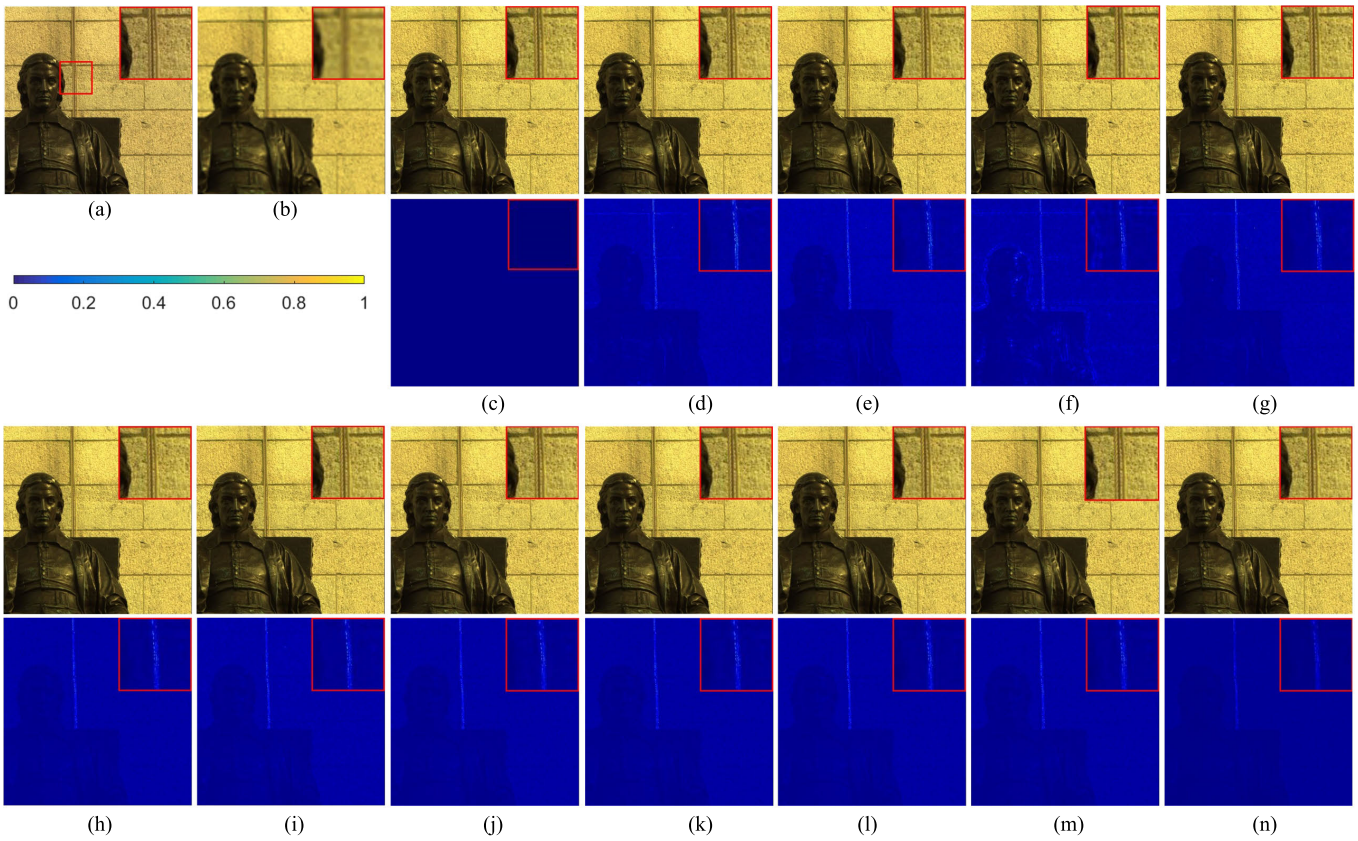


Fig. 7. Fusion results of different methods on the Harvard dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) CNMF. (e) HySure. (f) GLP-HS. (g) CSU. (h) SpfNet. (i) SSR-Net. (j) MHF-Net. (k) PSRT. (l) MSST-Net. (m) 3DT-Net. (n) Proposed SDGU-Net.

TABLE I
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS ON THE CAVE DATASET

Methods	PSNR	SAM	UIQI	SSIM	ERGAS
CNMF	45.4782	2.8711	0.9975	0.8517	1.6329
HySure	45.0399	4.5643	0.9980	0.8418	1.7005
GLP-HS	46.0922	2.1823	0.9985	0.8815	1.5542
CSU	43.6666	3.0246	0.9972	0.8541	2.3116
SpfNet	47.0318	2.0568	0.9991	0.8837	1.2882
SSR-Net	45.5955	3.7195	0.9979	0.8686	1.6147
MHF-Net	47.6287	2.3065	0.9991	0.8833	1.2297
PSRT	48.0599	1.9298	0.9992	0.8801	1.1345
MSST-Net	47.2415	2.0502	0.9992	0.8828	1.1939
3DT-Net	48.2349	**1.8068**	**0.9994**	0.8788	1.1007
SDGU-Net	**48.3968**	1.8916	**0.9994**	**0.8942**	**1.0681**

D. Results on the Harvard Dataset

Fig. 7 shows source images, GT, and the results of the different fusion methods. The visual results are placed in the first and third rows. Moreover, we select an interesting region and magnify it for further perception. The corresponding absolute error maps and the magnified error regions are also given in the second and fourth rows in Fig. 7. From the error maps, one can observe that the results of traditional methods include larger reconstruction errors when compared with some DNN-based methods, such as SpfNet, PSRT, MSST-Net, and 3DT-Net. Among DNN-based methods, the errors in the SSR-Net result cannot be ignored. For our proposed SDGU-Net, its result approximates the GT better, which is validated by the smaller errors in its corresponding absolute error map.

We compute the average values of all methods on the test image pairs, and the details are shown in Table II. The highest PSNR of the SDGU-Net result implies that the proposed method improves the spatial details in the fused image better. Besides, SDGU-Net also has better behavior in terms of UIQI and ERGAS. The best SAM and SSIM values come from 3DT-Net and MSST-Net, respectively, while the proposed SDGU-Net achieves suboptimal results in these two metrics. Thus, SDGU-Net has a better overall performance in terms of spatial and spectral information.

E. Results on the Pavia Dataset

In this part, we present the fusion results of our proposed SDGU-Net and the nine compared methods on the Pavia dataset. We select one LR HS and HR MS image pair for visualization, and Fig. 8 shows the fusion results and their corresponding absolute error maps. In addition, one region containing buildings is emphasized by a red box and magnified for a more direct comparison. In Fig. 8, one can observe that the spatial details are lost in the building and vegetation areas of the results of traditional methods, such as CNMF, GLP-HS, and CSU. For SpfNet and PSRT results, the errors in vegetation areas are suppressed well, but some large errors

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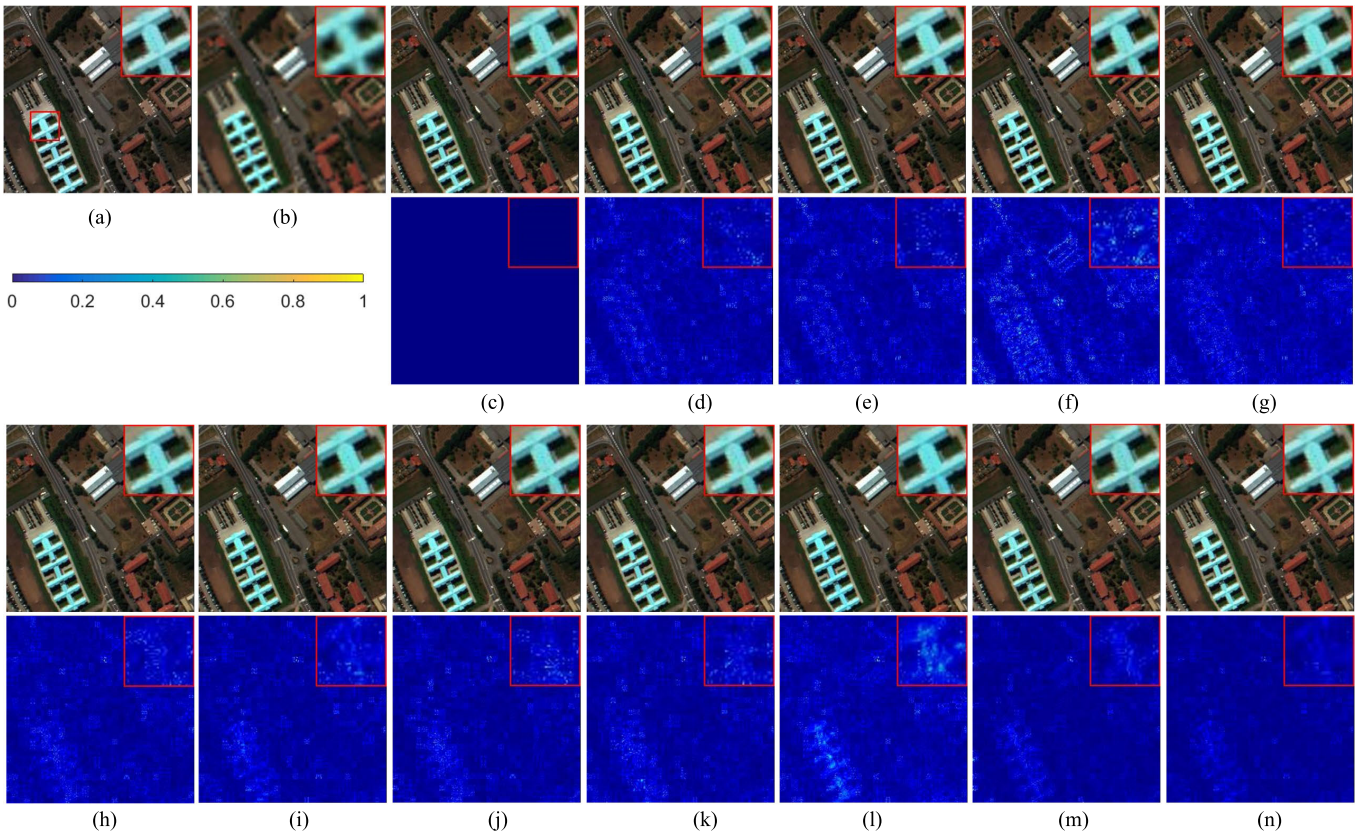


Fig. 8. Fusion results of different methods on the Pavia dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) CNMF. (e) HySure. (f) GLP-HS. (g) CSU. (h) SpfNet. (i) SSR-Net. (j) MHF-Net. (k) PSRT. (l) MSST-Net. (m) 3DT-Net. (n) Proposed SDGU-Net.

TABLE II
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS ON THE HARVARD DATASET

Methods	PSNR	SAM	UIQI	SSIM	ERGAS
CNMF	40.6286	2.2318	0.9920	0.9713	1.3850
HySure	39.8974	2.4576	0.9906	0.9661	1.5860
GLP-HS	39.8368	2.2775	0.9920	0.9696	1.3944
CSU	40.5064	2.2515	0.9921	0.9712	1.3779
SpfNet	42.0600	2.0525	0.9939	0.9746	1.1787
SSR-Net	41.1414	2.2865	0.9913	0.9711	1.4424
MHF-Net	41.8268	2.1019	0.9928	0.9738	1.3942
PSRT	42.0335	2.0894	0.9940	0.9742	1.1375
MSST-Net	42.0780	2.0574	0.9937	**0.9750**	1.1927
3DT-Net	42.085	**2.0466**	0.9939	0.9739	1.1339
SDGU-Net	**42.2121**	2.0481	**0.9942**	0.9748	**1.1122**

TABLE III
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS ON THE PAVIA DATASET

Methods	PSNR	SAM	UIQI	SSIM	ERGAS
CNMF	41.7426	1.9143	0.9983	0.9666	1.2681
HySure	41.7756	1.9985	0.9984	0.9626	1.2892
GLP-HS	38.7083	2.3998	0.9959	0.9539	1.7387
CSU	39.8805	2.1471	0.9965	0.9591	1.5456
SpfNet	44.0482	1.5766	0.9978	0.9677	1.0510
SSR-Net	46.1764	1.3628	0.9985	0.9735	0.8116
MHF-Net	45.0998	1.5023	**0.9994**	0.9680	0.9724
PSRT	44.5921	1.5271	0.9989	0.9696	1.5271
MSST-Net	41.0323	2.2156	0.9979	0.9502	2.2156
3DT-Net	44.2135	1.4020	0.9977	0.9703	1.0432
SDGU-Net	**46.2356**	**1.3218**	0.9985	**0.9742**	**0.8085**

still can be found in the building areas. The result shows that MSST-Net has some large reconstruction errors, which may be due to the weak generalization ability of the network. From the error maps, one can conclude that our proposed SDGU-Net demonstrates better approximation ability than other compared methods.

Table III lists the quantitative results of different methods on the Pavia dataset. The MHF-Net achieved the best UIQI. Meanwhile, it is easy to see that the proposed SDGU-Net produces the best values of the other four metrics. The good performance of the proposed SDGU-Net is because the unfolding architectures are derived from the graph regularized optimization.

F. Results on the DFC2018 Houston Dataset

To more fully illustrate the effectiveness of our proposed SDGU-Net, we again compared the reconstruction results of all methods on the DFC2018 Houston dataset. Fig. 9 depicts the visualization effect of fused images obtained through different methods and displays them in the first and third

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TABLE IV
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS ON THE DFC2018 HOUSTON DATASET

Methods	PSNR	SAM	UIQI	SSIM	ERGAS
CNMF	41.7917	1.1102	0.9981	0.9652	0.8142
HySure	42.1318	1.2274	0.9985	0.9518	0.8718
GLP-HS	39.2007	1.2439	0.9962	0.9588	0.9638
CSU	40.9320	1.2296	0.9977	0.9594	0.8537
SpfNet	50.9112	0.6587	**0.9998**	**0.9761**	0.3888
SSR-Net	50.1218	0.7328	0.9996	0.9729	0.4315
MHF-Net	46.9977	0.8663	0.9996	0.9669	0.5641
PSRT	50.5800	0.6915	0.9997	0.9740	0.3911
MSST-Net	49.5606	0.7464	0.9997	0.9732	0.4220
3DT-Net	50.1111	0.6531	0.9997	0.9751	0.3762
SDGU-Net	**51.3698**	**0.6511**	**0.9998**	**0.9761**	**0.3666**

rows together with the source images and GT. Meanwhile, the corresponding absolute error maps are shown in the second and fourth rows. Also, enlarge meaningful areas and mark them with red boxes. We can see that the spectral distortion and spatial artifacts are more severe in the fused images acquired by the traditional method. Among the DNN-based methods, SpfNet, SSR-Net, and PSRT have better error suppression. Furthermore, 3DT-Net also obtains a better overall performance with its reasonable unfolding method. However, MHF-Net cannot preserve spatial and spectral information and suffers from significant reconstruction errors. From the absolute error maps, although some DNN-based methods achieve advanced results, they are still inferior to SDGU-Net in edge and texture information extraction. Therefore, the proposed SDGU-Net has higher fusion quality on the DFC2018 Houston dataset.

Table IV shows the numerical metrics, with the best values in bold. As can be seen in Table IV, our proposed method performs best on all metrics. Also, SpfNet achieves the best UIQI and SSIM. This proves that our proposed SDGU-Net has better overall capability than other compared methods. Besides, it can be seen that DNN-based methods are significantly better to traditional ones on the Houston dataset. But, on other datasets, DNN-based methods may appear slightly better than traditional methods. This may be because the model can be fully trained when the dataset is large. On the contrary, when the dataset is small, traditional methods perform more stably. Furthermore, the effectiveness of these methods depends on the complexity of scenarios in datasets. For relatively simple scenarios, the differences between DNN-based methods and traditional methods in dealing with linear problems are not significant, resulting in close results. However, for complex scenarios, DNN-based methods have obvious advantages, as they can better solve complicated mapping problems and fit complex data. In summary, DNN-based methods have better overall performance.

G. Ablation Study

In the proposed SDGU-Net, the features are extracted by the cascaded spectral and spatial graph unfolding modules.

TABLE V
ABLATION STUDY OF DIFFERENT MODULES ON THE CAVE DATASET

Modules	PSNR	SAM	UIQI	SSIM	ERGAS
w/o spectral graph unfolding module	46.8191	2.0067	0.9993	0.8706	1.1899
w/o spatial graph unfolding module	47.8518	1.9855	0.9993	0.8919	1.1313
w/o feature fusion module	43.2181	2.8624	0.9981	0.8688	1.8503
w/o feature condense module	46.8028	2.0441	0.9992	0.8695	1.2138
w/o feature fusion module and feature condense module	44.3955	2.6677	0.9984	0.8757	1.6593
Complete SDGU-Net	**48.3968**	**1.8916**	**0.9994**	**0.8942**	**1.0681**

Then, these features are combined by feature fusion modules and further enhanced by the feature condense module for the reconstruction of the desired HR HS image. In the experiment part, they are ablated to analyze the influence of these modules. Specifically, spectral and spatial graph unfolding modules are replaced by vanilla convolution blocks. The feature condense module is removed, and a vanilla convolution block is introduced to adjust the number of channels in feature maps. Moreover, we verify the effectiveness of our proposed graph regularization by removing both the feature fusion module and the feature condense module and replacing them with vanilla convolution blocks.

Fig. 10 shows the fusion results of the different settings mentioned above. The absolute error maps in the second row of Fig. 10 imply that the ablation of different modules has a negative influence on the fusion results. For instance, when spectral or spatial graph unfolding modules are removed, the reconstruction errors in texture areas become larger. In addition, the removal of feature fusion modules has a larger impact on the fused image. It proves that sufficient fusion in feature space can further boost the quality of the fused image. In order to demonstrate that our adopted graph regularization approach has basic feature extraction capability, we eliminate both the feature fusion module and the feature condense module. From Fig. 10(f), it can be seen that when we only use the graph unfolding module, the overall information of the image is appropriately preserved. After introducing the feature fusion module and feature condense module, the detailed information is supplemented to obtain better fusion results, as shown in Fig. 10(h). Besides, Table V reports the evaluation results of different settings in ablation experiments. The performance in terms of all metrics is consistent with the results in Fig. 10. Finally, the complete SDGU-Net is used for comparison.

Subsequently, we also ablate spectral GCNs and spatial GCNs to better demonstrate the role of graph representation in blocks. The following experiments are performed on the CAVE dataset. Fig. 11 shows the visual results and absolute error maps for different configurations. We first ablate the spectral GCN to verify its ability to capture SBC. It can be seen that the spectral distortion in Fig. 11(d) is more serious than that in Fig. 11(f), and the color information is not well preserved. This indicates that our proposed spectral GCN can retain spectral information better. Then, we remove the spatial GCN from the network to explore its effect in acquiring SNS. From Fig. 11(e), we can observe that eliminating the spatial

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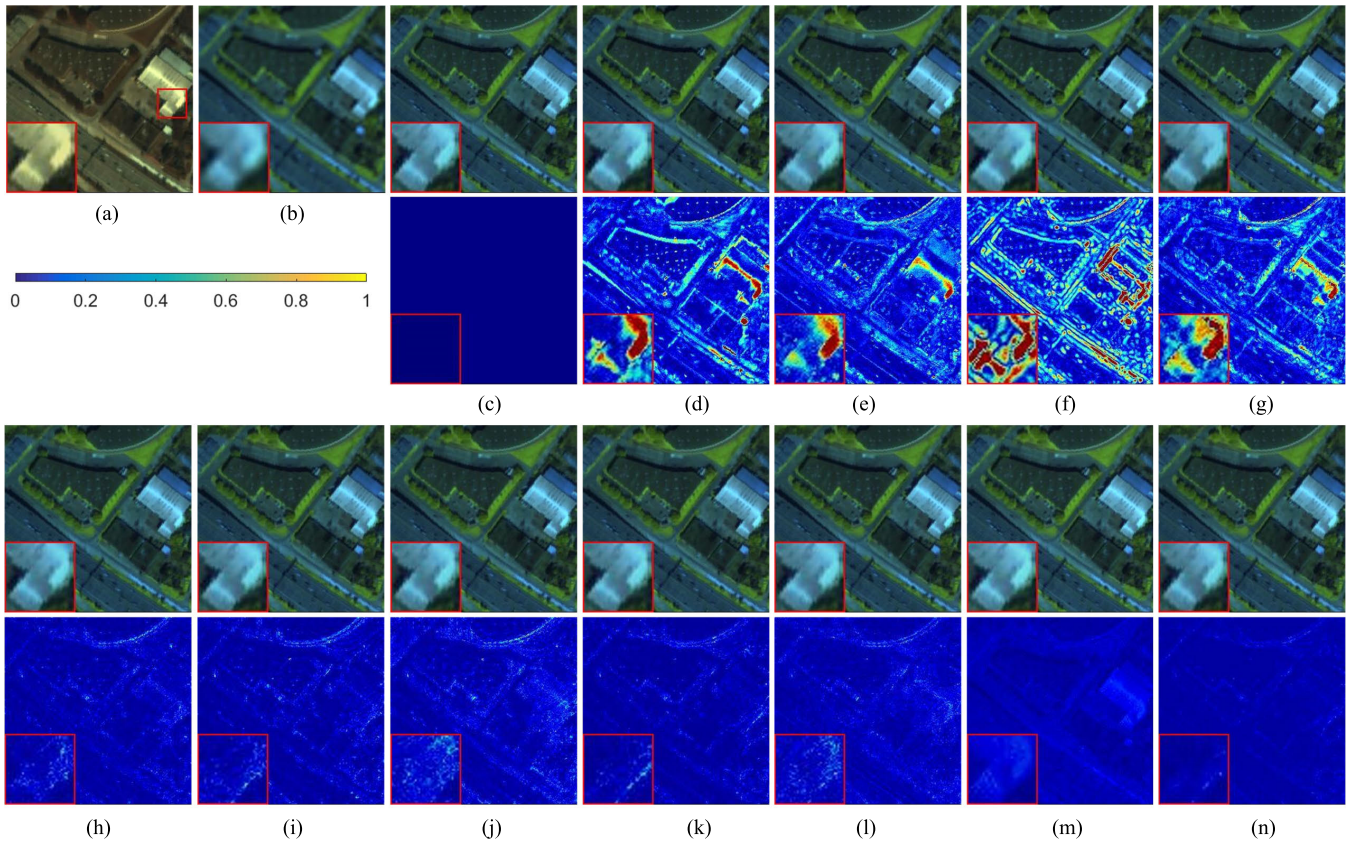


Fig. 9. Fusion results of different methods on the DFC2018 Houston dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) CNMF. (e) HySure. (f) GLP-HS. (g) CSU. (h) SpfNet. (i) SSR-Net. (j) MHF-Net. (k) PSRT. (l) MSST-Net. (m) 3DT-Net. (n) Proposed SDGU-Net.

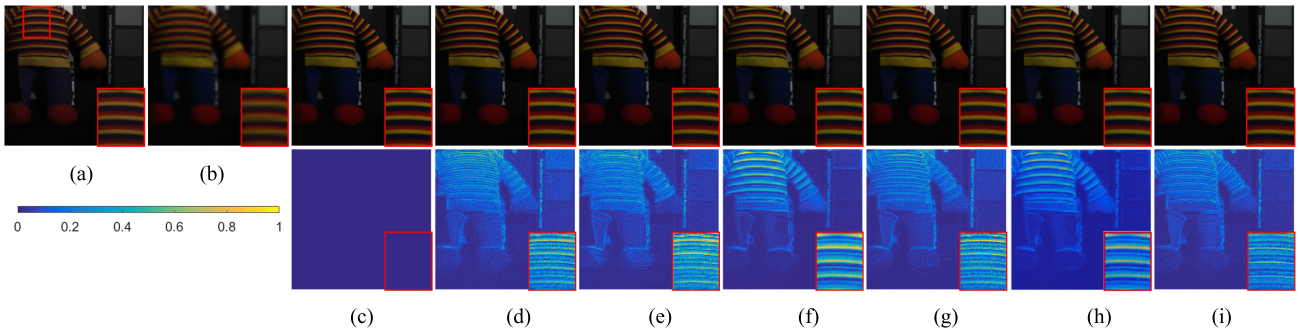


Fig. 10. Ablation study of different modules on the CAVE dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) w/o spectral graph unfolding module. (e) w/o spatial graph unfolding module. (f) w/o feature fusion module. (g) w/o feature condense module. (h) w/o feature fusion module and feature condense module. (i) Complete SDGU-Net.

GCN introduces spatial artifacts compared with the complete SDGU-Net. Therefore, this spatial GCN has good properties for mining spatial details. Meanwhile, Table VI records the numerical metrics corresponding to Fig. 11, with the best results labeled in bold. In Table VI, we can also see that designing spectral GCN and spatial GCN has a positive effect on obtaining spectral and spatial information.

TABLE VI
ABLATION STUDY OF SPECTRAL GCN AND SPATIAL GCN ON THE CAVE DATASET

GCNs	PSNR	SAM	UIQI	SSIM	ERGAS
w/o spectral GCN	48.0064	1.9463	0.9993	0.8906	1.1217
w/o spatial GCN	48.0436	1.9512	0.9993	0.8913	1.1072
Complete SDGU-Net	**48.3968**	**1.8916**	**0.9994**	**0.8942**	**1.0681**

H. Network Architecture

This part describes the influence of the number of stages on the CAVE dataset. In our proposed SDGU-Net, the depth increases when more stages are cascaded, and meanwhile, the number of parameters also increases. Fig. 12 illustrates the fused HR HS images of SDGU-Net with different numbers of stages. From Fig. 12, one can see that the SDGU-Net with one stage suffers from some spatial artifacts in the areas containing

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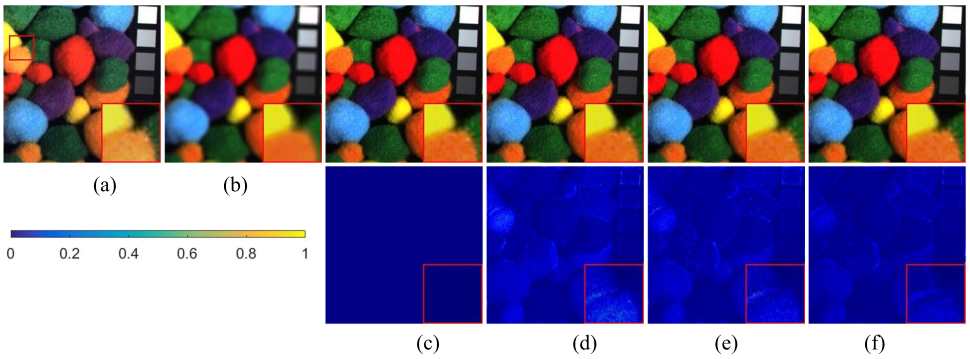


Fig. 11. Ablation study of spectral GCN and spatial GCN on the CAVE dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) w/o spectral GCN. (e) w/o spatial GCN. (f) Complete SDGU-Net.

TABLE VII
INFLUENCE OF THE NUMBER OF STAGES ON THE CAVE DATASET

#Stages	PSNR	SAM	UIQI	SSIM	ERGAS	#Para.
1	48.3149	1.9052	0.9993	0.8926	1.0924	**610k**
2	**48.3968**	**1.8916**	**0.9994**	**0.8942**	**1.0681**	740k
3	46.7540	2.0128	0.9993	0.8697	1.2191	869k
4	46.7353	2.0376	0.9992	0.8701	1.2240	998k

TABLE VIII
INFLUENCE OF THE WEIGHT SHARING ON THE CAVE DATASET

Strategies	PSNR	SAM	UIQI	SSIM	ERGAS	#Para.
Shared weights	48.1945	1.9179	0.9993	**0.8944**	1.1294	**610k**
Independent weights	**48.3968**	**1.8916**	**0.9994**	0.8942	**1.0681**	740k

TABLE IX
INFLUENCE OF PATCH SIZE ON THE CAVE DATASET

Patch Sizes	PSNR	SAM	UIQI	SSIM	ERGAS
4×4	48.3876	1.8949	0.9993	0.8931	1.0917
8×8	**48.3968**	**1.8916**	**0.9994**	**0.8942**	**1.0681**
12×12	48.1100	1.9313	0.9993	0.8904	1.1247
16×16	48.2470	1.9059	0.9993	0.8910	1.0901

textures. For the SDGU-Net consisting of three or four stages, their results have inferior performance in the flat areas, such as the squares in the top-right corners in these images. Thus, we adopt two stages to extract the features in source images by the spectral and spatial graph unfolding modules.

Table VII provides the metric values of SDGU-Net with different numbers of stages. The numbers of parameters of these settings are also given in Table VII. The values in this table also suggest that the best values are achieved when spectral and spatial graph unfolding modules are cascaded two times. The reason for this may be that the learning ability of the proposed SDNU-Net with one stage is limited, and it cannot learn the spatial and spectral information in HS images efficiently. For the SDGU-Net made up of three or four stages, the number of parameters is larger, and these parameters cannot be trained sufficiently on the constructed datasets. Meanwhile, deeper networks are typically harder to train due to the well-known problem of vanishing gradients, which leads to degraded results.

Besides, we also investigate the influence of the weight-sharing strategy on the fused results. Here, the SDGU-Net with two stages is considered. In the weight-sharing strategy, the weights in these two stages are the same. Fig. 13 displays the results produced by the SDGU-Net with different strategies. It can be seen that larger errors appear in the fusion result when the weight-sharing strategy is adopted by the proposed SDGU-Net. The reason for this may be that the representation ability is limited when the weights in different modules are shared. Table VIII provides the results of different strategies. Although the best SSIM value is from the weight-sharing strategy and its number of parameters is also smaller, the best values of the other four metrics are acquired by the SDGU-Net with independent weights. The quantitative evaluations in Table VIII indicate that the SDGU-Net with independent weights is better. So, the weights in the two stages of the SDGU-Net are not shared.

I. Influence of Patch Size

In addition, we investigate the influence of patch size in the spatial graph unfolding module on fusion results and explore how to utilize SNS more efficiently. To verify its influence on SDGU-Net, we evaluate the performance of the proposed SDGU-Net with different patch sizes, such as 4×4, 8×8, 12×12, and 16×16. Fig. 14 shows the experimental results and absolute error maps on the CAVE dataset. From Fig. 14(e) it can be seen that better spectral fidelity and spatial details are achieved for a patch size of 8×8. In other words, small or large patch sizes bring larger spectral corruptions and spatial errors. Their corresponding results are shown in Fig. 14(d), (f), and (g). Meanwhile, Table IX records the experimental results evaluated quantitatively on the CAVE dataset, and the best results are marked in bold. It is obvious that when the patch size is 8×8, the best results are obtained among all indicators. Finally, we set the patch size as 8×8 to achieve the best fusion performance of SDGU-Net.

J. Parameter Analysis

In (2) and (6), the regularization parameters α and β are introduced to control the contributions of spectral and spatial

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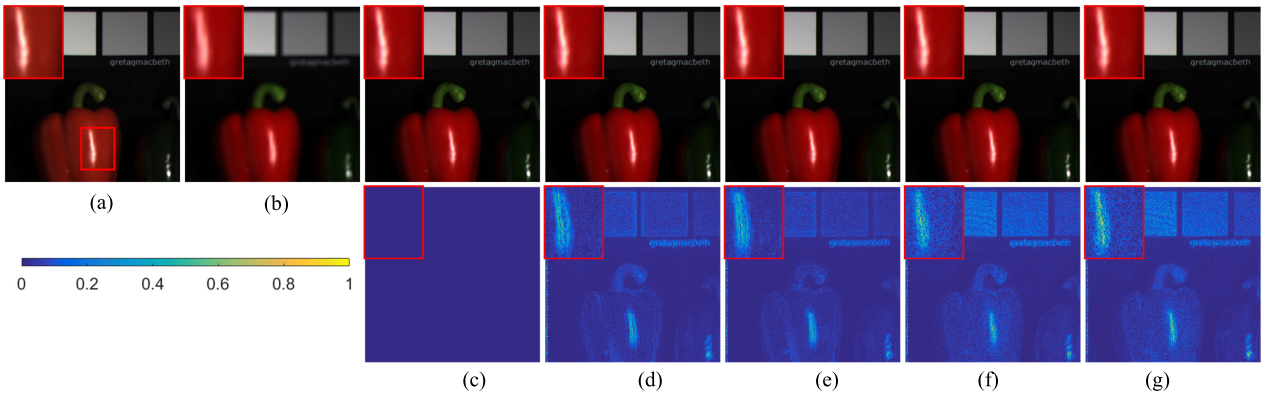


Fig. 12. Influence of the number of stages on the CAVE dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) One stage. (e) Two stages. (f) Three stages. (g) Four stages.

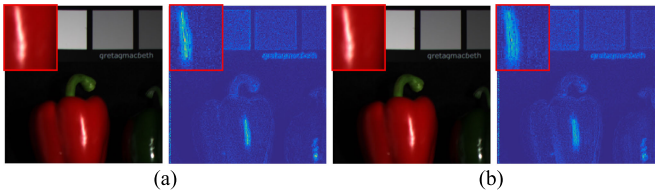


Fig. 13. Influence of weight sharing on the CAVE dataset. (a) Two stages with shared weights. (b) Two stages with independent weights.

graph constraints. This part explores the influence of the two parameters on the fusion results by settings α and β to different combinations of the following four values 0.1, 0.3, 0.5, and 0.8. Also, we try to set both of them as learnable parameters. Table X records the results of the quantitative assessment for different combinations of trade-off parameters. The best SSIM is obtained when we set α and β to 0.1. It can be seen that setting α and β to 0.1 and 0.5 gives the best results for SAM and ERGAS. Meanwhile, the highest PSNR and UIQI are achieved when both α and β are 0.5. Moreover, SSIM and ERGAS also achieve suboptimal results. Similarly, UIQI achieves the best results on two other combination approaches. Subsequently, we set α and β as learnable parameters but this setting does not show enough competitiveness compared with other fixed-value combinations. The reason for this may be that the learnable α and β may suffer from dramatical variations during training, and the spatial and spectral similarity priors cannot be constrained stably. In conclusion, SDGU-Net has better overall performance when α and β are 0.5. Therefore, we set α and β to 0.5.

In addition, the upsampled LR HS and HR MS images are also combined by different weights μ and ρ to explore their influence on fusion results. Fig. 15 shows the fused images of different weights on one image pair from the CAVE dataset. From the HR MS and LR HS images in Fig. 15, we can know that there are abundant color and edge information. When we set μ and ρ to 0.2 and 0.8, some large errors arise in the error maps of the obtained result, especially the contours of the objects in the result. A similar phenomenon also occurs in the results of Fig. 15(e), (g), and (h). Compared with the results of other settings, the reconstruction errors are smaller when μ and ρ are set as 0.5. Table XI provides quantitative results for different weights. We can see that when μ and ρ are too small or too large, the spectral and spatial information is not preserved well enough to produce satisfactory fusion results. The best values are produced by setting μ and ρ to 0.5.

TABLE X
INFLUENCE OF TRADE-OFF PARAMETERS ON THE CAVE DATASET

α, β	PSNR	SAM	UIQI	SSIM	ERGAS
0.1, 0.1	48.2620	1.8888	0.9993	**0.8946**	1.1048
0.1, 0.3	48.0175	1.9555	0.9993	0.8897	1.1183
0.1, 0.5	48.3191	**1.8870**	0.9993	0.8922	**1.0664**
0.1, 0.8	48.1114	1.9152	0.9993	0.8920	1.1153
0.3, 0.1	48.1417	1.8965	0.9992	0.8926	1.2272
0.3, 0.3	48.1219	1.9495	0.9993	0.8924	1.1198
0.3, 0.5	48.1374	2.0033	0.9993	0.8897	1.1073
0.3, 0.8	48.0898	1.9156	0.9993	0.8909	1.1189
0.5, 0.1	48.3455	1.8892	**0.9994**	0.8918	1.1045
0.5, 0.3	47.8559	2.0382	0.9992	0.8903	1.1786
0.5, 0.5	**48.3968**	1.8916	**0.9994**	0.8942	1.0681
0.5, 0.8	48.2230	1.9602	0.9993	0.8906	1.1145
0.8, 0.1	48.0486	1.9351	0.9993	0.8910	1.1287
0.8, 0.3	48.0269	1.9134	0.9992	0.8917	1.2017
0.8, 0.5	48.2270	1.9224	0.9993	0.8924	1.0976
0.8, 0.8	48.3034	1.9285	**0.9994**	0.8940	1.0841
learnable	48.0597	1.9414	0.9993	0.8902	1.1052

TABLE XI
INFLUENCE OF UPSAMPLING WEIGHTS ON THE CAVE DATASET

μ, ρ	PSNR	SAM	UIQI	SSIM	ERGAS
0.2, 0.8	48.0821	1.9219	0.9993	0.8900	1.1069
0.3, 0.7	48.0765	1.9475	0.9993	0.8905	1.1431
0.5, 0.5	**48.3968**	**1.8916**	**0.9994**	**0.8942**	**1.0681**
0.7, 0.3	48.3855	1.9075	0.9993	0.8924	1.1060
0.8, 0.2	48.1535	2.2637	0.9993	0.8838	1.1328

K. Analysis of Interpolation Methods

In this section, we explore the effect of different upsampling methods on the fusion results. Three upsampling methods are considered for comparison, namely, bilinear interpolation,

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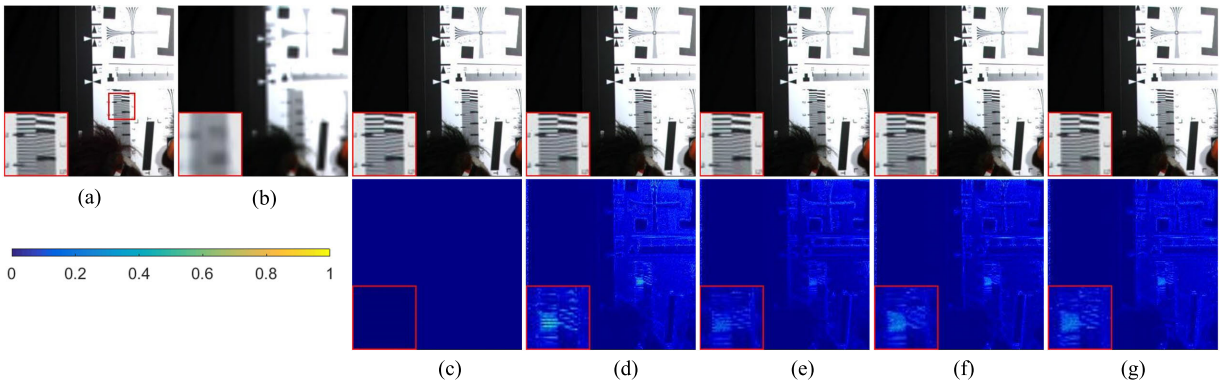


Fig. 14. Fusion results of SDGU-Net with different patch sizes on the CAVE dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) 4×4. (e) 8×8. (f) 12×12. (g) 16×16.

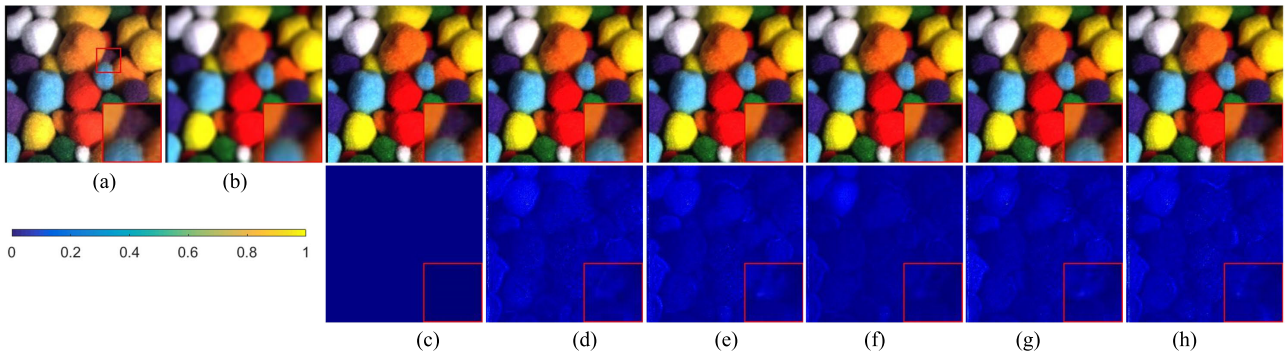


Fig. 15. Fusion results of SDGU-Net with different upsampling weights on the CAVE dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) $\mu = 0.2, \rho = 0.8$. (e) $\mu = 0.3, \rho = 0.7$. (f) $\mu = 0.5, \rho = 0.5$. (g) $\mu = 0.7, \rho = 0.3$. (h) $\mu = 0.8, \rho = 0.2$.

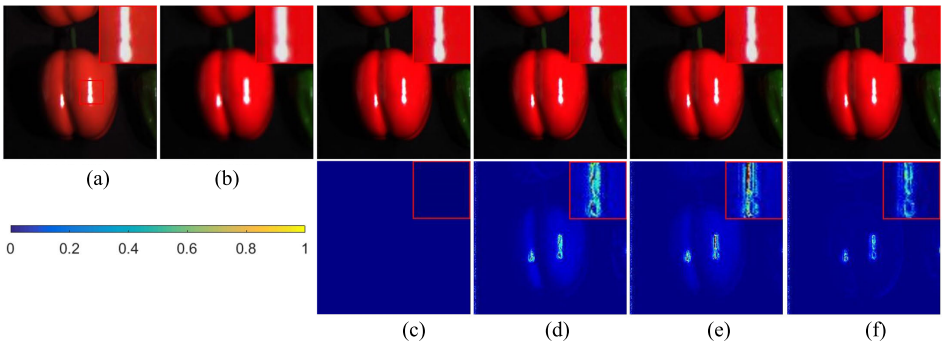


Fig. 16. Fusion results of SDGU-Net with different interpolation methods on the CAVE dataset. (a) HR MS image. (b) LR HS image. (c) GT. (d) Bilinear. (e) Nearest. (f) Bicubic.

nearest interpolation, and bicubic interpolation. Fig. 16 shows the visualization results, including pseudo-color images and absolute error maps. It can be easily seen that the bicubic interpolation method has smaller reconstruction errors and clearer edge and texture information. The quantitative results on the CAVE dataset are shown in Table XII, and the best results are marked in bold. We find that SDGU-Net achieves the best results on four metrics using bicubic interpolation. In addition, nearest interpolation performs best on SAM and UIQI metrics. Overall, we choose bicubic interpolation as the upsampling method for fusion.

TABLE XII
INFLUENCE OF INTERPOLATION METHODS ON THE CAVE DATASET

Interpolation	PSNR	SAM	UIQI	SSIM	ERGAS
bilinear	48.0758	1.9249	0.9993	0.8909	1.1262
nearest	48.2867	**1.8727**	**0.9994**	0.8898	1.1010
bicubic	**48.3968**	1.8916	**0.9994**	**0.8942**	**1.0681**

L. Analysis of Model Complexity

It is also known that the model size is an important metric for evaluating DNN-based methods. Therefore, we compare the computational complexity of SDGU-Net and other DNN-based methods on the CAVE dataset, including parameters, floating-point operations (FLOPs), and runtime in this section. From Table XIII, we can see that the computational complexity of the proposed SDGU-Net is only higher than that of SSR-Net and PSRT. This may be because SDGU-Net needs to be unfolded into multiple stages and uses GCNs. However,

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TABLE XIII
MODEL SIZE, FLOPS, AND RUNTIME OF THE PROPOSED SDGU-NET AND OTHER DNN-BASED METHODS ON THE CAVE DATASET

Methods	SpfNet	SSR-Net	MHF-Net	PSRT	MSST-Net	SDGU-Net
#Para.	1204k	**26k**	1862k	247k	33409k	740k
FLOPs	103.50G	**1.70G**	151.9G	16.8G	746.38G	53.21G
Runtime	21.4h	**4.8h**	168.2h	34.6h	34.1h	33.6h

compared with the unfolding-based DNN methods, such as SpfNet and MHF-Net, our proposed SDGU-Net has a smaller model size and is accompanied by better reconstruction results. At the same time, SDGU-Net has a competitive performance in terms of runtime. Considering the quality of fusion, the proposed SDGU-Net has better overall performance.

VI. CONCLUSION

In this article, we propose the SDGU-Net for the fusion of LR HS and HR MS images. We regularize the reconstruction of the fused image by the graph Laplacian matrices containing the SBC and SNS priors, and the proximal gradient descent-based iterative algorithms are deduced to solve the models. Considering the good performance of the model-driven DNNs, we unfold the iterative algorithms as the proposed SDGU-Net, in which spectral and spatial graph unfolding modules are cascaded for the learning of the SBC and SNS in feature space. In spectral and spatial graph unfolding modules, the similarity among different nodes is captured by GCNs and used to enhance the SBC and SNS in the extracted features. Then, these features are merged by feature fusion modules, in which the spectral and spatial information is further improved by residual learning. Finally, we use a feature condense module to generate the fusion result. We conduct experiments on the CAVE, Harvard, and Pavia datasets, and the experimental results show that the spectral and spatial information is preserved efficiently by our proposed SDGU-Net. In our method, the spectral response should be prepared in advance. For future work, we will develop more efficient architectures to achieve the blind fusion of LR HS and HR MS images. Besides, some hyperparameter modules will also be considered to calculate the regularization parameters according to the content in LR HS and HR MS images.

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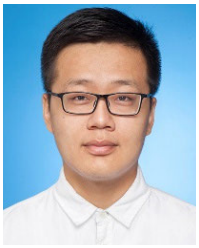
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